

Tutorial 01. Perspective Projection

Solutions

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Convention. $O = (0, 0, 0)$, $f > 0$, and scene points have $Z > 0$ unless stated otherwise. On the virtual image plane, the principal point is the origin and x, y are parallel to X, Y , with y upward:

$$\pi(X, Y, Z) = (fX/Z, fY/Z).$$

Numerical examples below use $f = 1$. Core Q1–Q7 match the website. E1–E2 are optional extensions.

<https://pinhole-camera-lab.chenrui-fan.chatgpt.site/>

Q1. Projected points of a line are collinear

Question. Show that the projectable points of a straight 3D line are collinear in the image. State what happens for a line through the camera centre.

Solution.

A 3D line that does not contain O , together with O , determines a plane. For a nonzero normal n , every point $P = (X, Y, Z)$ on the line satisfies

$$n_x X + n_y Y + n_z Z = 0.$$

For $Z \neq 0$, divide by Z and substitute the projection equations:

$$\boxed{n_x x + n_y y + f n_z = 0.}$$

If the line has projectable points, this describes a 2D line containing their images. It does not assert that the images cover that entire line.

For example, $P(t) = (t, 1, 2 + t)$ with $t > -2$ gives $x = t/(2 + t)$ and $y = 1/(2 + t)$, so $x + 2y = 1$. The point $(1, 0)$ is approached only as depth tends to infinity.

If the 3D line passes through O , its projectable points lie on one camera ray and collapse to a single image point. Points with $Z = 0$ have no finite projection; a line entirely in that plane has none.

Q2. A shared point remains shared

Question. If two 3D lines share a projectable point, show that its image belongs to both projected sets. Must the two image lines have a unique crossing?

Solution.

Suppose $P \in L_1 \cap L_2$ and P is projectable. Its unique image $p = \pi(P)$ belongs to both projected sets:

$$\pi(P) \in \pi(L_1) \cap \pi(L_2).$$

For $L_1(t) = (1 + t, 1, 2)$ and $L_2(s) = (1, 1 + s, 2)$, the common point $(1, 1, 2)$ maps to $(1/2, 1/2)$.

The image lines can coincide, and a line through O can collapse to a point. Thus shared-point membership is the general conclusion; a unique crossing of two distinct image lines is not guaranteed.

Q3. The converse is false

Question. Show by a concrete counterexample that intersecting images do not imply that the corresponding 3D lines intersect.

Solution.

Take

$$L_1(t) = (t, 0, 1), \quad L_2(s) = (0, s, 2).$$

The 3D lines are disjoint because their depths differ. Their images are

$$\pi(L_1(t)) = (t, 0), \quad \pi(L_2(s)) = (0, s/2),$$

which intersect at the origin. In particular, the distinct finite points $(0, 0, 1)$ and $(0, 0, 2)$ have the same image. Projection is not one-to-one: it loses position along a camera ray.

This is a finite-depth counterexample. The common vanishing-point limit of parallel lines is a different construction, discussed in Q7. Occlusion may hide a point in a physical opaque scene, but does not alter this geometric argument.

Q4. Angles are not preserved in general

Question. Show by a counterexample that angles are not preserved in general under perspective projection.

Solution.

Let two segments run from $P = (0, 0, 2)$ to $P + V$ and $P + W$, where

$$V = (1, 0, 1), \quad W = (1, 1, -1).$$

All segment points are in front of the camera. Since $V \cdot W = 1 + 0 - 1 = 0$, their 3D angle is 90° . Their endpoint images are

$$\pi(P) = (0, 0), \quad \pi(P + V) = (1/3, 0), \quad \pi(P + W) = (1, 1).$$

The image directions $(1, 0)$ and $(1, 1)$ have cosine $1/\sqrt{2}$, hence angle 45° . This counterexample disproves general angle preservation. Some angles are preserved in special configurations; equal metric scales on the image axes are assumed.

Q5. Lengths depend on depth

Question. Show that lengths are not preserved in general. State the assumptions under which a vertical segment has image height $h = fH/Z$.

Solution.

For a vertical segment with endpoints (X, Y_1, Z) and (X, Y_2, Z) at a common positive depth, let $H = |Y_2 - Y_1|$. Then

$$h = |y_2 - y_1| = f \frac{|Y_2 - Y_1|}{Z} = \boxed{\frac{fH}{Z}}.$$

Two unit-height segments at $Z = 2$ and $Z = 4$ therefore have image heights $1/2$ and $1/4$ when $f = 1$. Equal 3D lengths need not have equal image lengths. If endpoints have different depths, project them separately; one common Z cannot be used. In a purely mathematical extension to $Z < 0$, the magnitude denominator is $|Z|$.

Q6. The horizon and its back-projected plane

Question. For a level camera, determine the 3D plane through the camera centre that corresponds to the horizon. Distinguish this plane from the actual ground below the camera.

Solution.

For a level camera with vertical Y , the horizon is $y = 0$. Its back-projected rays lie in the plane

$$\Pi_h = \{(X, 0, Z) : X, Z \in \mathbb{R}\}.$$

or, equivalently, $\text{span}\{(1, 0, 0), (0, 0, 1)\}$. This plane passes through O and is parallel to the ground; it is not the ground below the camera.

For example, on ground $Y = -1$, the lines $(\pm 1, -1, Z)$ have images $(\pm 1/Z, -1/Z)$ and approach $(0, 0)$ as $Z \rightarrow \infty$. Other ground-parallel directions give other horizon points.

More generally, let N be the ground-plane normal in camera coordinates. Horizon directions (x, y, f) are perpendicular to N , so

$$N_x x + N_y y + f N_z = 0.$$

Camera orientation changes the horizon position. Translating the ground to a parallel plane does not change its vanishing line.

Q7. Parallel lines and vanishing-point limits

Question. Show that parallel 3D lines may converge to a vanishing point in the image. Derive the point for direction V when $V_z \neq 0$, and discuss $V_z = 0$.

Solution.

For $P(t) = P_0 + tV$, projection gives

$$p(t) = f \left(\frac{X_0 + tV_x}{Z_0 + tV_z}, \frac{Y_0 + tV_y}{Z_0 + tV_z} \right).$$

If $V_z \neq 0$, divide each numerator and denominator by t and take the unbounded parameter direction that keeps depth positive:

$$p_\infty = f(V_x/V_z, V_y/V_z).$$

The starting point disappears, so parallel lines share this limit. For $P_a(t) = (a + t, 0.5, 2 + t)$ with $a = -1, 1$, both images approach $(1, 0)$. Their second coordinates are positive at every finite positive depth; neither line reaches that limiting point at finite depth.

If $V_z = 0$, there is no finite vanishing point. Provided they have projectable points, the image lines are parallel or coincident. The 3D lines remain parallel throughout.

E1. Perspective projection of an opaque sphere (optional)

Question. Determine the perspective silhouette of an opaque sphere of radius $R > 0$ and centre $S = (S_x, S_y, S_z)$, assuming $S_z > R$ so the entire sphere is in front of the camera. When is its image a circle?

Solution.

Let the sphere have centre $S = (S_x, S_y, S_z)$, radius $R > 0$, and $S_z > R$. The entire sphere is in front of the camera and the camera lies outside it. Its silhouette is formed by tangent rays, not by projecting an arbitrary great circle.

For image point (x, y) , write $q = (x, y, f)$ and the corresponding camera ray as λq , $\lambda > 0$. Intersections with the sphere satisfy

$$\|\lambda q - S\|^2 = R^2,$$

or

$$\lambda^2 \|q\|^2 - 2\lambda(S \cdot q) + \|S\|^2 - R^2 = 0.$$

Tangency means a repeated root, so the discriminant vanishes:

$$(S \cdot q)^2 = (\|S\|^2 - R^2)\|q\|^2.$$

This is a quadratic equation in x, y , the silhouette conic. Under $S_z > R$ it is a bounded ellipse (including the circular special case). The opaque sphere projects to its filled interior; equivalently, forward intersecting rays have nonnegative discriminant and $S \cdot q > 0$.

To see why the boundary is an ellipse, put $s = (S_x, S_y)^T$, $u = (x, y)^T$, and $A = \|S\|^2 - R^2$. The equation becomes

$$u^T(AI - ss^T)u - 2fS_z s^T u + f^2(A - S_z^2) = 0.$$

The eigenvalues of $AI - ss^T$ are A and $A - \|s\|^2 = S_z^2 - R^2$, both positive. The two eigenvalues are equal exactly when $s = 0$, so, for metric image axes, the ellipse is a circle when the sphere centre is on the optical axis.

For $S = (0, 0, d)$ with $d > R$, the equation reduces to

$$(d^2 - R^2)(x^2 + y^2) = f^2 R^2,$$

giving image radius

$$r = \frac{fR}{\sqrt{d^2 - R^2}}.$$

Only for $R \ll d$ is $r \approx fR/d$. Removing the wholly-in-front assumption requires a separate analysis of the conic, the forward-ray condition and the visible branch; the bounded-ellipse claim above is deliberately scoped to $S_z > R$.

E2. Orthographic projection of a sphere (optional)

Question. Under orthographic projection along Z , what is the image of the same sphere?

Solution.

Orthographic projection along Z maps (X, Y, Z) to (X, Y) . A point belongs to the sphere if

$$(X - S_x)^2 + (Y - S_y)^2 + (Z - S_z)^2 = R^2.$$

There exists a real Z precisely when

$$\boxed{(X - S_x)^2 + (Y - S_y)^2 \leq R^2.}$$

Thus the opaque sphere projects to a disk of radius R , centred at (S_x, S_y) , and its silhouette is a circle. A separate image-unit scale, if introduced, scales this radius accordingly.